Plasma levels of amino acids correlate with motor fluctuations in parkinsonism.

Seven patients with Parkinson's disease who experienced severe motor fluctuations in response to levodopa were studied in detail with relation to the effect of dietary protein on their motor function. The levodopa dose for each patient was not changed during the period of study, and no other antiparkinsonian drugs were used. Regular and high-protein diets resulted in a marked elevation in the plasma concentrations of large neutral amino acids (LNAAs) that are known to compete with levodopa for transport across the blood-brain barrier. Despite elevated plasma levodopa levels, all patients with elevated LNAA levels experienced parkinsonian symptoms. When the amino acid level dropped while plasma levodopa levels were elevated, patients experienced relief of these symptoms. On a low-protein diet, LNAA levels remained low and all patients were consistently dyskinetic throughout the day, even though the mean plasma levodopa levels were somewhat lower than when the patients consumed a high-protein diet. A redistribution diet that is virtually protein free until supper and then unrestricted until bedtime is tolerated by patients because this simple manipulation permits near-normal daytime motor function.